CHART: Curriculum-guided Hierarchy-Aware Relation classification via prompt Tuning

Anonymous Author(s)

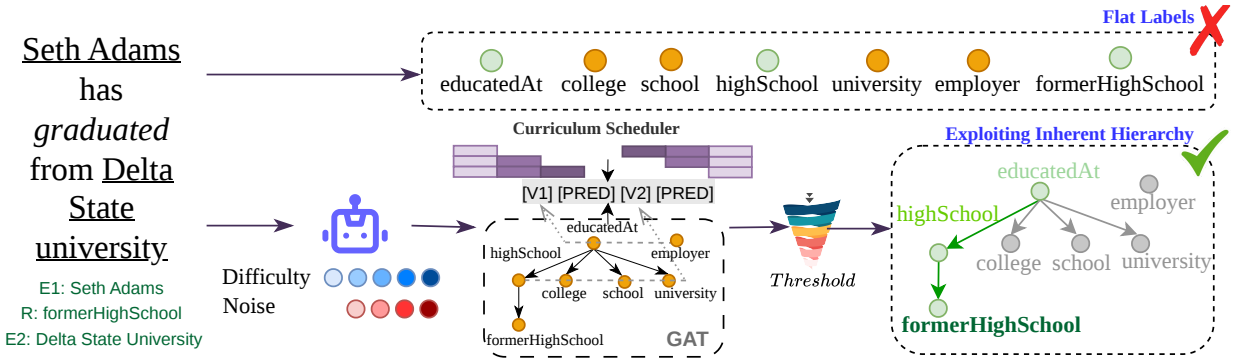

Figure 1: The proposed CHART framework. By combining LLM-guided curriculum learning, GAT injected prompt tuning, and dynamic thresholding, our model extracts consistent hierarchical paths, overcoming the limitations of flat-label classification.

Abstract

Relation Classification (RC) is a fundamental task of information extraction. Most of the existing approaches formulate RC task with relatively small number of relation labels and treat the labels as a flat set. Consideration of large number of relations requires the use of synthetic, LLM-generated datasets. This introduces varying levels of instance difficulty and noise along with evident hierarchical nature of relation labels. Current models largely overlook such challenges. Addressing these challenges, this work proposes *Curriculum-guided Hierarchy-Aware Relation classification via prompt Tuning* (*CHART*). CHART introduces a reference-free LLM-as-a-Judge module to automatically estimate the difficulty and noise of training instances. A curriculum scheduler utilizes this to organize data into optimal learning phases. To explicitly capture semantic dependencies, we design a layer-wise hierarchy-aware prompt enhanced by Graph Attention Network (GAT)-injected template embeddings. Finally, the model is optimized using a zero-bounded multi-label cross-entropy loss for robust, layer-wise consistent predictions, along with a cross-validation-based dynamic thresholding strategy for adaptive inference. Extensive experiments on a synthetic dataset (AmalREC) and two human-annotated benchmarks (TACRED and Re-TACRED) demonstrate the effectiveness of `CHART` by consistently outperforming 12 existing baselines. Additionally, comprehensive ablation studies over 12 curriculum scheduling strategies reveal that combining difficulty expanding and noise annealing yields the most stable and robust performance.

CCS Concepts

• **Computing methodologies** → **Information extraction**.

Keywords

Hierarchical Relation Classification, Prompt Tuning, Curriculum Learning

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1 Introduction

Relation Extraction (RE) identifies semantic links between entities in text, typically via a two-step pipeline: entity extraction followed by Relation Classification (RC). RC predicts the relationship between a given entity pair from a predefined label set, serving as a core component for knowledge base construction, question answering, and information retrieval. Most of the existing work [3, 4, 16, 22, 35] treat relation labels as a flat set, ignoring the inherent structure among them. In practice, many relation labels naturally form a hierarchy or fine-grained sub-relations. As illustrated in Figure 1, a given sentence can correspond to multiple plausible relation labels such as *educatedAt*, *highSchool*, and *formerHighSchool*. However, these relations are not independent; instead, they are organized in

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a hierarchical structure where *formerHighSchool* represents a more fine-grained and semantically specific relation under broader categories. Accurately identifying such fine-grained relations requires the model to respect the hierarchical dependencies among labels. While prior works have explored hierarchical relation extraction [9, 31, 37], their approaches remain limited. Despite these efforts, existing methods still suffer from several limitations such as (i) many approaches rely on specialized architectures that are not easily compatible with modern pre-trained language models.(ii) they often fail to explicitly enforce consistency across different levels of the hierarchy during prediction. (iii) the role of data quality such as instance difficulty and noise is largely overlooked, which can significantly affect model learning and generalization. As a result, effectively integrating hierarchical structure with robust training strategies remains an open challenge in relation extraction. Several hierarchical-RE frameworks do not explicitly integrate parent–child dependencies into the prompt/template representations, nor do they combine curriculum-guided exposure with a structured, zero-bounded multi-label loss that enforces hierarchical consistency at the layer level. To overcome these limitations, we propose CHART, a novel framework that introduces an LLM-as-a-Judge-guided curriculum scheduler, a hierarchy-aware prompt with GAT-injected template embeddings, and a zero-bounded multi-label cross-entropy loss applied per hierarchy layer, enabling more robust and semantically coherent hierarchical relation predictions. In summary, the main contributions of this work are as follows:

- We propose `CHART`, a Curriculum-guided Hierarchy-Aware Relation classification via prompt Tuning method that formulates a fully supervised hierarchical pipeline with curriculum based data scheduling, improving stability and consistency. To the best of our knowledge, this is the first work on hierarchical relation classification.
- We introduce a reference-free evaluation mechanism using Large Language Models (LLMs) to automatically estimate the difficulty and noise of training instances, enabling adaptive curriculum construction.
- We conduct a comprehensive comparative analysis of `CHART` against 12 state-of-the-art baselines across 3 benchmark relation extraction datasets.
- We conduct an extensive ablation study over 12 different curriculum scheduling strategies, analyzing various combinations of difficulty- and noise-based expanding and annealing schemes across multiple datasets, providing detailed insights into their impact on performance.

Our proposed framework of four key components. First, we employ an LLM-as-a-Judge module to estimate the difficulty and noise level of each training instance, enabling automatic assessment of data quality. Next, we construct a curriculum scheduler that organizes the training data into shards based on these estimates and presents them in a structured learning order using expanding or annealing strategies. Subsequently, we design a hierarchy-aware prompt that reformulates the relation classification task into a masked prediction problem with layer-wise constraints aligned with the relation hierarchy. To further capture dependencies among relation labels, we incorporate a graph-based hierarchy injection mechanism that encodes parent-child relationships directly into the prompt embeddings. Finally, the model is optimized using a zero-bounded multi-label objective, which enforces separation between positive and negative labels while maintaining consistency across hierarchical levels.

Beyond modeling challenges, training data quality critically impacts relation extraction. We evaluate our method against 12 baselines on 3 diverse datasets, including the synthetic AmalREC[21]. While synthetic data improves diversity and scale, it introduces noise, ambiguity, and inconsistent semantics, alongside large hierarchical label spaces that hinder learning. We address this by leveraging label hierarchies and a curriculum-guided training strategy based on automated difficulty and noise estimation, providing a robust testbed for handling heterogeneous data.

2 Related Work

Relation Classification: Relation extraction (RE) aims to identify semantic relations between entities in text. Existing approaches can be broadly categorized into pipeline-based methods, which first detect entities and then classify relations, and joint methods that predict entity–relation triplets simultaneously [16]. In this work, we focus on the second step of pipeline-based methods, known as Relation Classification (RC), which assumes the entity pairs are already provided [3, 4, 35]. Most existing RC approaches treat relation labels as a flat set, ignoring inherent hierarchical structures among them. However, many relations naturally form hierarchies, where fine-grained relations can be abstracted into higher-level categories [22]. Deep learning methods have shown strong performance for RC, but they often require large annotated datasets, leading to the exploration of distant supervision and few-shot learning approaches. Among these, context encoding methods based on architectures such as CNNs, RNNs, attention mechanisms, and pre-trained language models play a key role in capturing sentence-level semantics for accurate relation prediction [16].

Prompt Tuning: Prompt tuning [17], [19] provides a new paradigm to stimulate the relation information of PLMs by bridging the format gap between the pre-training tasks and the downstream RE tasks. Works like [10], [26], [27], [13] show that prompt learning can effectively leverage the knowledge encoded in the PLMs. There are predominantly two kinds of prompt tuning, i.e., hard prompting and soft prompting, as shown in Figure 5. Both soft prompt and hard prompt do not take labels into account until prediction, and PLM views all candidate words as equal. Previous works suggest that incorporating label dependency instead of modeling them as flat classification is essential for alleviating the label imbalance [5]. While Sainz's work [25] attempted reformulating RC as textual entailment with handcrafted verbalisations, such methods still require considerable expert knowledge and are not easily adaptable to new datasets or relation types. GenPT [8] has proposed reformulating RC as a text infilling task. This approach could eliminate rigid prompt restrictions. KnowPrompt [4] could solve the problem of using multiple tokens by constructing a verbalizer for each label through having a single virtual answer word within the PLM vocabulary space that could fully represent relation labels.

Curriculum Learning (CL): As a model training strategy for non-convex optimization, curriculum learning [2] advocates that models learn in the order of easier to more difficult data, mimicking the basic idea of gradual human learning as they learn curriculum [18].

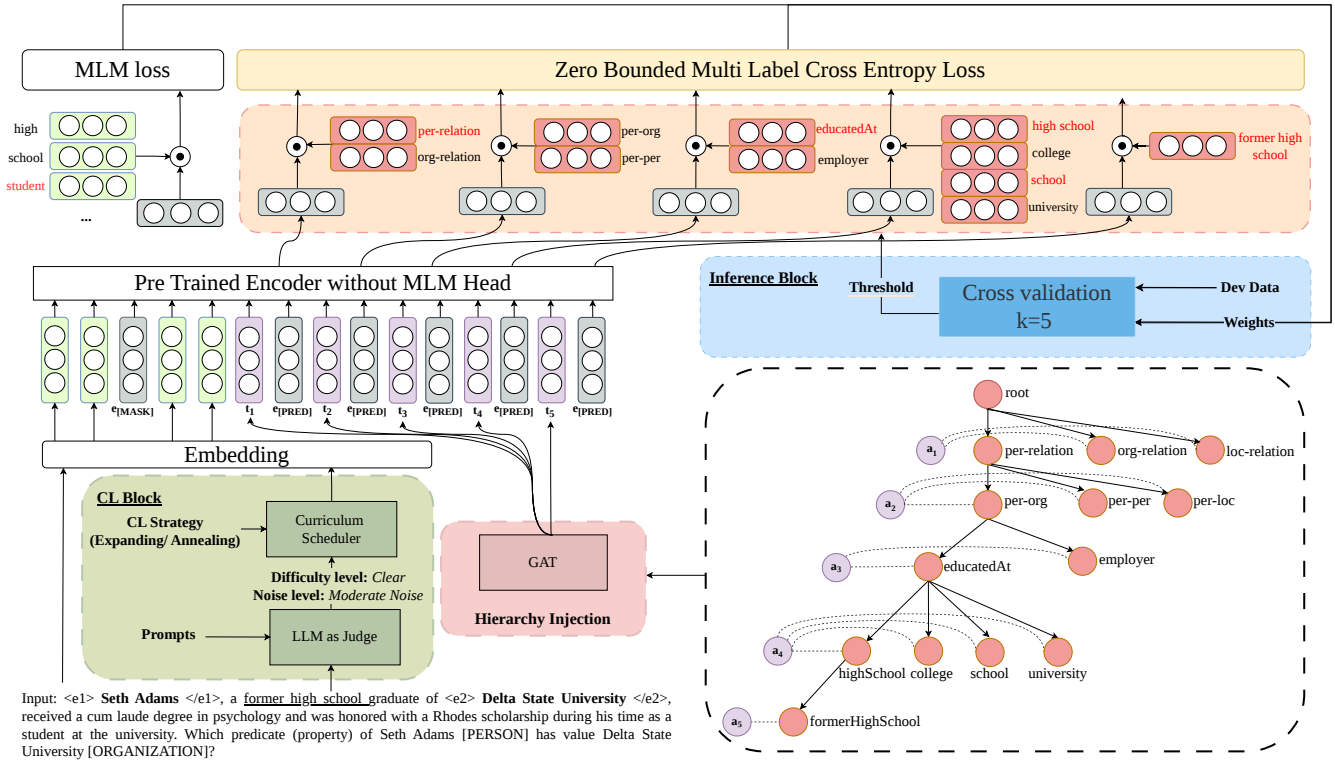

Figure 2: Architecture of CHART. The input sentence is first embedded and evaluated by an LLM-based judge to assign difficulty and noise levels, which guide a curriculum scheduler for training. A pre-trained encoder with hierarchy-aware prompt tokens processes the input, while a GAT module injects label hierarchy information into the prompt representations. The model jointly optimizes MLM loss and a zero-bounded multi-label cross-entropy loss to predict layer-wise relation labels. During inference, a cross-validated threshold is applied to ensure consistent hierarchical predictions across the label tree.

Existing curriculum learning methods can be classified as data-based, task-based, or model-based according to the object of their application. We focus on the first. It mainly focuses on starting with easy samples and gradually increasing complexity and knowledge during training. The definition of CL has changed over the years, such as starting with most relevant tasks first [24] or training with imbalanced to balanced training subsets [33] or learning from easy to representative samples in sequence [12].

3 Methodology

We propose CHART (**C**urriculum-guided **H**ierarchy-**A**ware **R**elation classification via prompt **T**uning), a novel framework that integrates hierarchy-aware prompt tuning with curriculum learning for hierarchical relation classification. The proposed approach aims to enhance learning stability while enforcing hierarchical consistency in relation prediction, leading to more robust and semantically coherent predictions, as illustrated in Figure 2.

We define the hierarchical relation classification task as follows. Given an input sentence x, a pair of entities $(E_1, E_{1\text{Type}})$ and $(E_2, E_{2\text{Type}})$, where E_1 and E_2 denote the entity mentions and $E_{1\text{Type}}$ and $E_{2\text{Type}}$ denote their corresponding entity types, and a relation label hierarchy $H = (Y, \mathcal{E})$, where Y is the set of relation labels (nodes) and $\mathcal{E}$ encodes the parent-child dependencies among them, the objective is to learn a function $f(x, E_1, E_2)$ that predicts a relation label $y \in Y$ while respecting the hierarchical structure imposed by H.

Our framework consists of four main key components. First, an LLM-as-a-Judge module estimates the difficulty and noise of training instances. Second, a curriculum scheduler organizes data into shards and presents them using structured learning strategies. Third, a hierarchy-aware prompt reformulates relation classification as a layer-wise masked prediction task, further enhanced by graph-based hierarchy injection. Finally, the model is trained using a zero-bounded multi-label objective to ensure hierarchically consistent predictions, complemented by a cross-validation-based dynamic thresholding strategy for adaptive decision-making during inference.

3.1 LLM as a Judge

We employ large language models (LLMs) as a reference-free evaluation mechanism, commonly referred to as *LLM-as-a-Judge* [40], to estimate the difficulty and noise level of training instances. Specifically, the input tuple (x, E_1, E_2) is provided to the LLM through a structured prompt. The prompt includes: (i) a formal definition of the relation classification task, (ii) a set of predefined difficulty and noise categories, and (iii) instructions for assigning the most appropriate category for each dimension. Here, the *difficulty* level measures how challenging it is to infer the relation between the entities from the

Table 1: Predefined difficulty and noise levels used for annotation and curriculum learning.

Level	Description
Difficulty Levels	
Clear	The relation is clearly expressed or directly paraphrased in the sentence.
Contextual	The relation is not directly stated but can be inferred using clues, synonyms, or rewording in context.
Implicit	requires deeper reasoning, understanding subtle cues, and choosing among multiple plausible relations.
Knowledge Dependent	Requires external factual or commonsense knowledge to infer the correct property.
Impossible	The sentence lacks sufficient information, is too ambiguous, or the RDF triple (E_1, y, E_2) is incompatible with the sentence due to entity mention errors or relation mismatch.
Noise Levels	
Clean	The sentence is grammatically sound, well-aligned with the relation, and clearly expresses the subject-object association.
Minor Noise	Slight grammatical issues, alignment problems, or irrelevant but non-conflicting content are present, but the relation can still be inferred.
Moderate Noise	The sentence contains factual inaccuracies, loosely fits the relation, or includes conflicting or unclear clauses.
Severe Noise	The sentence is misaligned with the relation tuple, contains contradictory or incoherent information, or lacks a discernible connection between entities.

sentence, while the *noise* level captures the quality and reliability of the sentence with respect to the relation tuple. We define the difficulty dimension across five levels and the noise dimension across four levels, as described in Table 1. This structured prompting strategy reduces bias and improves consistency in evaluation. The complete prompt details are provided in Appendix A. The constructed prompt is passed to the LLM, which outputs the predicted difficulty and noise levels along with a short justification. Samples identified as ambiguous or hallucinated are automatically assigned the *Impossible* difficulty and *Severe Noise* level. Table 9 shows some examples of noise and difficulty judgements on AmalREC dataset.

To evaluate the robustness and model-agnostic nature of the proposed prompting strategy, we conduct an agreement analysis across multiple LLMs such as Llama v3, Mistral, SmolL, Zephyr, etc. Specifically, we sample a statistically significant subset of the dataset and obtain difficulty and noise annotations from different LLMs. Table 7 reports the pairwise agreement scores between models, where the disagreement between any pair of LLMs remains below approximately 30%. Based on this observation, we adopt a majority voting scheme to determine the final difficulty and noise labels for each instance. The overall distribution of the assigned difficulty and noise levels across the dataset is presented in Table 8.

3.2 Curriculum Scheduler

Based on the estimated difficulty and noise levels obtained from the LLM-as-a-Judge module, we construct a curriculum scheduler to organize the training process. Specifically, each training instance (x, E_1, E_2) is associated with a difficulty score and a noise score, which are used to guide the ordering of samples during training. Curriculum learning involves presenting training samples in a structured and meaningful order, rather than randomly, to facilitate more effective learning. This approach mimics natural learning processes, where models gradually learn from simpler to more complex examples [2]. To implement this, we partition the training data into multiple shards based on the estimated difficulty and noise levels. The training process is then carried out in multiple phases, where each phase utilizes a specific subset of these shards. This progressive exposure helps the model learn robust representations while mitigating the impact of noisy samples. We explore two fundamental curriculum scheduling strategies to organize the training phases (an illustrative diagram is provided in Appendix Figure 4):

(1) Expanding: Training begins with the easiest and least noisy shards. In subsequent phases, progressively more difficult and noisier shards are introduced, allowing the model to gradually adapt to complex instances.

(2) Annealing: Training starts with all shards in the initial phase. In later phases, the easier and cleaner shards are gradually removed, forcing the model to focus on harder and noisier examples. The shards are randomly shuffled within each phase.

We further explore different combinations of difficulty-based and noise-based scheduling under expanding & annealing strategies in section 6.

3.3 Hierarchy-aware Prompt

Following the curriculum scheduler, which organizes training samples based on difficulty and noise, we design a hierarchy-aware prompt to explicitly model the dependencies among relation labels during prediction. To incorporate the hierarchical structure of relation labels, we adopt a layer-wise prompt design that aligns with the label hierarchy. Since the hierarchy can be represented as a tree, we construct the prompt template based on its depth.

Given an input sentence x, entity pair $(E_1, E_{1\text{Type}})$ and $(E_2, E_{2\text{Type}})$, and a relation hierarchy H with depth L, we construct a prompt template that explicitly aligns with the hierarchical levels.

The template is defined as:

```
"[CLS] x Based on the sentence, which
predicate (property) of E1 [E1_TYPE]
has value E2 [E2_TYPE]? [V1] [PRED] [V2]
[PRED] ... [VL] [PRED] [SEP]".
```

Here, each `[PRED]` token corresponds to a specific level in the hierarchy, enabling layer-wise prediction. $[V_i]$ denotes a learnable virtual template token associated with the i-th level. The prompt is encoded into embedding vectors to obtain the input representation below equation 1.

$$T = [x_1, x_2, \ldots, x_N, t_1, e_P, \ldots, t_L, e_P], \tag{1}$$

where $X = [x_1, x_2, \ldots, x_N]$ represents the embeddings of the input sentence along with the hard prompt tokens, t_i denotes learnable template embeddings, and e_P is the embedding of the `[PRED]` token, initialized using the `[MASK]` embedding. A pre-trained language model encodes T to produce hidden representations as equation 2

$$H = [h_1, h_2, \ldots, h_N, h_{t_1}, h_P^1, \ldots, h_{t_L}, h_P^L], \tag{2}$$

where h_P^i corresponds to the hidden state of the i-th `[PRED]` token and is responsible for predicting labels at the i-th level of the hierarchy.

To represent relation labels, we construct learnable label embeddings v_i by averaging their corresponding token embeddings. To enforce hierarchical constraints, each $\texttt{[PRED]}_i$ is restricted to predict labels only from the i-th layer of the hierarchy. This layer-wise decomposition enables the model to capture dependencies across levels and alleviates label imbalance by distributing predictions across multiple slots. A verbalizer is used to map the hidden states of [PRED] tokens to relation labels. Formally, for the m-th layer, the verbalizer Verb_m is defined as equation 3.

$$\text{Verb}_m(y_i) = \begin{cases} v_i, & y_i \in N_m \\ \varnothing, & \text{otherwise} \end{cases} \tag{3}$$

where N_m denotes the set of labels at the m-th layer, and $\varnothing$ indicates that labels outside this layer are not considered. This design enforces layer-wise prediction consistency and allows the model to explicitly leverage the hierarchical structure during inference.

3.4 Injection of Hierarchy

While the hierarchy-aware prompt captures layer-wise prediction, it does not explicitly model the dependencies among relation labels defined by the hierarchy $H = (Y, \mathcal{E})$, where $\mathcal{E}$ encodes the parent-child relationships. To address this limitation, we inject hierarchical information directly into the prompt template embeddings using a Graph Attention Network (GAT) [14]. Given a node u (one relation of Y) at the k-th GAT layer, the information aggregation is defined:

$$g_u^{(k+1)} = \text{ReLU}\left(\sum_{v \in N(u) \cup \{u\}} \frac{1}{c_u} W^{(k)} g_v^{(k)}\right) \tag{4}$$

where $N(u)$ denotes the neighbors of node u, c_u is a normalization constant, and $W^{(k)} \in \mathbb{R}^{d_m \times d_m}$ is a trainable weight matrix.

To encode layer-wise hierarchical information, we augment the relation hierarchy graph by introducing L virtual nodes $a_1, a_2, \ldots, a_L$, where each node a_i is connected to all relation label nodes at the i-th level of the hierarchy H. This design allows each virtual node to aggregate information not only from labels at the same level but also implicitly from lower levels through message passing. Each node u in the graph is initialized with a feature representation. For relation label nodes, we use their corresponding label embeddings v_i, while for virtual nodes a_i, we initialize them using the template embeddings t_i. After applying K layers of GAT, each virtual node a_i obtains an updated representation $g_{a_i}^K$, which captures the hierarchical dependencies among labels. We then inject this information into the prompt by updating the template embeddings using a residual connection in equation 5.

$$t_i' = t_i + g_{a_i}^K \tag{5}$$

The updated template embeddings t_i' replace the original t_i, thereby enabling the prompt to incorporate hierarchical relationships directly into the representation space.

3.5 Zero Bounded Multi Label Cross Entropy Loss

Binary cross-entropy loss (6) is commonly used in multiple binary classification problems.

$$L_{\text{BCE}} = -\sum_{i=1}^{C} \left(y_i \log(s_{y_i}) + (1 - y_i) \log(1 - s_{y_i})\right) \tag{6}$$

where s_{y_i} is the predicted sigmoid score (between 0-1) for output label y_i. We cannot use this because it ignores the connection between y_i's or ignores the hierarchical nature between them and treats them as a flat set. MLM's are inherently trained for multi-class (predicting a single class out of many classes) classification task which is optimized with the cross-entropy loss 7

$$L_{\text{CE}} = -\log \frac{e^{s_{y_t}}}{\sum_{i=1}^{C} e^{s_{y_i}}} = \log\left(1 + \sum_{i=1, i \neq t}^{C} e^{s_{y_i} - s_{y_t}}\right) \tag{7}$$

where y_t is the ground truth label for the input. The cross-entropy loss enforces the score of the ground truth label to be higher than those of all other classes, thereby implicitly capturing inter-class relationships during optimization.

To integrate these two different goals of multi-class and multi-label, we optimise that for all ground truth labels to have a score greater than all non-ground truth labels. This can be modelled by multi-label cross-entropy loss (MLCE) (8).

$$L_{\text{MLCE}} = \log\left(1 + \sum_{y_i \in N^n} \sum_{y_j \in N^p} e^{s_{y_i} - s_{y_j}}\right) \tag{8}$$

where N^p and N^n represent positive (ground truth) labels and negative (non-ground truth) labels, respectively. However, the prediction could be made for the entire set of labels during inference. To solve this, [29] introduced a new loss Zero Bounded Multi Label Cross Entropy loss (9) which fixes zero as the threshold in MLCE and optimizes that ground truth labels achieve a score greater than 0 and non-ground truth labels achieve a score less than 0.

$$\begin{aligned} L_{\text{ZMLCE}} &= \log\left(1 + \sum_{y_i \in N^n} \sum_{y_j \in N^p} e^{s_{y_i} - s_{y_j}} + \sum_{y_i \in N^n} e^{s_{y_i}} + \sum_{y_j \in N^p} e^{-s_{y_j}}\right) \\ &= \log\left(1 + \sum_{y_i \in N^n} e^{s_{y_i}}\right) + \log\left(1 + \sum_{y_j \in N^p} e^{-s_{y_j}}\right) \end{aligned} \tag{9}$$

This loss needs to be integrated with the hierarchy constraint. Hence we employ ZMLCE for every layer for layer-wise prediction. So formally, for m-th layer, the scores predicted by hidden state h_p^m, loss looks like this

$$L_{\text{ZMLCE}}^m = \log\left(1 + \sum_{y_i \in N_m^n} e^{s_{y_i}}\right) + \log\left(1 + \sum_{y_j \in N_m^p} e^{-s_{y_j}}\right) \tag{10}$$

where $s_{y_i} = v_i^T h_p^m + b_{im}$ and b_{im} is a learnable bias term. N_m^p and N_m^n are ground truth labels and non ground truth labels, respectively, of layer m. Overall the final loss can be described as the sum of ZMLCE loss for all L layers and MLM loss:

$$L_{all} = \sum_{m=1}^{L} L_{ZMLCE}^m + L_{MLM} \tag{11}$$

L_{MLM} is computed by randomly masking 15% of the text.

3.6 Finding Threshold using Cross Validation

During inference we observed that because of the noise in the dataset, sometimes the predicted scores for all the relation labels were negative, but the ground truth labels had a relatively less negative score.

Table 2: Dataset statistics

Dataset	Y	Depth	$\lvert y_i \rvert$	Train	Dev	Test
TACRED	42	4	1622	68124	22631	15509
Re-TACRED	40	4	1461.62	58465	19584	13418
AmalREC	224	5	673.49	150861	37716	15021

This led us to explore a mechanism for finding a dynamic threshold instead of 0. k-Cross-validation [15] using dev data is a common approach for such tasks. We evaluate using k-cross validation on dev data on a range of thresholds. We need to evaluate for all the negative range, but 0 and above should be taken care by the loss itself. The threshold that achieves the maximum average score on all k folds is the decided threshold for evaluating on the test set.

4 Experiments

Datasets and Evaluation Metrics: We experiment on TACRED [39], Re-TACRED [28], and AmalREC [21] datasets for evaluation. AmalREC is a synthetically generated dataset constructed through the amalgamation of multiple Large Language Models (LLMs), where the input tuples are sourced from the taxonomical hierarchy over DBpedia proposed by [22]. Inspired by this, we construct hierarchical label structures for TACRED and Re-TACRED by adapting the taxonomy proposed in [23, 30]. We use Micro-F1 as the primary evaluation metric to compare various experiments and baselines.

Baselines: We compare our method against 12 representative baselines approaches. *Supervised baselines* include standard fine-tuning of BERT [36] for relation classification using a sequence classification framework. *Prompt tuning baselines* consist of methods that leverage prompts to guide relation classification. KnowPrompt [4] introduces a knowledge-aware prompt tuning strategy that incorporates external knowledge into template design and answer construction. GenPT [8] reformulates relation classification as a generative infilling problem using prompt tuning, while PTR [10] constructs logic-based sub-prompts to model relational constraints. We also compare against HPT [34], a hybrid prompt tuning framework designed specifically for relation classification. *Graph-based baselines* include syntax-aware models that utilize dependency structures to capture relational patterns. AGGCN [7] employs Attention-Guided Graph Convolutional Networks over dependency graphs, while GCN over pruned trees [38] applies graph convolutional networks over pruned dependency trees to improve relation classification performance. Finally, we compare with five *LLM prompting baselines*, including (1) `meta-llama/Llama-3.2-3B`[1] [6], (2) `meta-llama/Llama-3.1-8B`[2] [6], (3) `mistralai/Mistral-7B-Instruct-v0.2`[3] [11], (4) `HuggingFaceH4/zephyr-7b-beta`[4] [32], and (5) `HuggingFaceTB/SmolLM3-3B`[5] [1]. All models are evaluated under zero-shot ($k = 0$) and few-shot ($k = 5$) prompting settings. The detailed implementation details of baselines are described in Appendix Section B. We conduct experiments to analyze the proposed method using both `bert-base-uncased` [36] and `roberta-base` [20] as the base architectures. The hyperparameters used in our proposed method are summarized in Table 4. All experiments are conducted on NVIDIA A100 GPUs (80GB and 40GB configurations).

[1] https://huggingface.co/meta-llama/Llama-3.2-3B
[2] https://huggingface.co/meta-llama/Llama-3.1-8B
[3] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2
[4] https://huggingface.co/HuggingFaceH4/zephyr-7b-beta
[5] https://huggingface.co/HuggingFaceTB/SmolLM3-3B

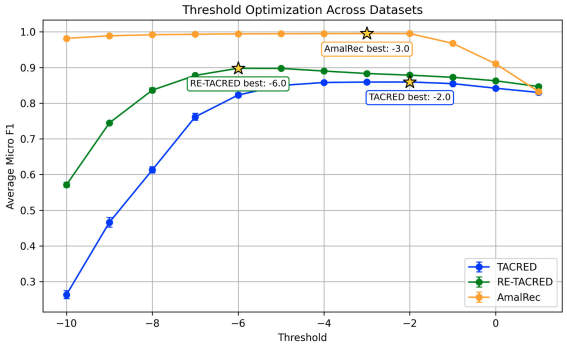

Figure 3: Micro F1 at various thresholds across datasets

5 Results

We present the main experimental results in Table 3, comparing `CHART` with a wide range of baselines across the TACRED, Re-TACRED, and AmalREC datasets using Micro-F1 as the primary evaluation metric. Overall, our proposed method achieves strong and consistent performance across both human-annotated datasets and noisy, large-scale synthetic data. In particular, `CHART` with RoBERTa achieves the best performance on TACRED (89.08) and AmalREC (86.26), outperforming all competing methods. On Re-TACRED, `CHART` remains competitive (89.50). While it trails behind KnowPrompt (91.29) on this specific dataset, KnowPrompt heavily relies on manually curated external knowledge and complex prompt engineering. In contrast, `CHART` leverages a unified framework that generalizes far better to complex, noisy datasets like AmalREC.

Our results validate the advantage of explicit hierarchical modeling over conventional flat-label classification. Compared to standard fine-tuning (BERT), `CHART` yields superior results on TACRED and AmalREC, and comparable performance on Re-TACRED. Among prompt-based baselines, GenPT and HPT show competitive performance with GenPT performing particularly well on Re-TACRED. However, `CHART` significantly outperforms these prompt-based methods on TACRED and the synthetically generated AmalREC. This demonstrates that our layer-wise prompting strategy, optimized with a zero-bounded multi-label cross-entropy loss, enforces semantic consistency across diverse relation types more effectively than standard text-infilling.

Furthermore, we observe that explicitly injecting parent-child label hierarchies via GAT is far more effective than relying solely on sentence-level syntactic trees. Graph-based approaches such as AGGCN and GCN over pruned trees perform substantially worse across all datasets (e.g., scoring only 52.99 and 73.96 on AmalREC), suggesting that syntactic dependency structures alone are insufficient for capturing complex semantic relationships. Similarly, zero-shot and few-shot ($k = 5$) LLM-based approaches (such as LLaMA, Mistral, and Zephyr) severely underperform compared to supervised and prompt-tuning methods. This indicates that, despite their strong generative capabilities, direct prompting of LLMs is inadequate for fine-grained hierarchical relation classification without task-specific training and structured objectives.

Table 3: Comparison of various RE/RC extraction baselines and prompting-based LLM approaches. Few-shot prompting uses $k = 5$ examples. Best results are shown in bold. Curriculum strategy followed for both RoBERTa and BERT models of CHART is (noise annealing + difficulty expanding).

Category	Method	TACRED	Re-TACRED	AmalREC
Supervised	Fine-Tuning BERT	88.70	91.00	82.12
Prompt Tuning	KnowPrompt [4]	70.39	**91.29**	82.49
	GenPT (RoBERTa) [8]	69.72	90.33	84.37
	PTR (reversed) [10]	72.40	89.78	68.43
	HPT [34]	86.75	85.63	76.73
Graph-Based	AGGCN [7]	67.46	75.20	52.99
	GCN pruned trees [38]	64.40	78.70	73.96
LLM Prompting	BART ($k = 0$)	18.92	28.31	34.04
	LLaMa 3.1 8B ($k = 0$)	31.49	43.78	56.58
	LLaMa 3.1 8B ($k = 5$)	23.65	39.41	62.26
	Mistral-7B ($k = 5$)	53.95	53.09	57.72
	Zephyr-7B ($k = 5$)	77.52	71.21	50.82
Our Method	CHART (BERT)	87.63	88.20	84.22
	CHART (RoBERTa)	**89.08**	89.50	**86.26**

Table 4: Hyperparameter settings used for training & evaluation.

Hyperparameter	Value
Graph encoder	Single-layer GAT
Batch size	4
Optimizer	Adam
Base learning rate	3×10^{-5}
Backbone LR	0.5× base LR
Other modules LR	1.0× base LR
Gradient clipping	1.0 (max norm)
LR scheduler	ReduceLROnPlateau
LR decay factor	0.5
LR patience	2 epochs
Minimum learning rate	1×10^{-7}
Number of epochs	5
Evaluation frequency	Every epoch
Random seed	3
Threshold range	[-10, 1]
Cross-validation folds	5

Finally, using RoBERTa consistently improves performance over BERT, showing our framework seamlessly scales with stronger contextual representations. Ultimately, the robust performance of `CHART` stems from three core components: (i) hierarchy-aware prompting that enforces structured predictions, (ii) graph-based hierarchy injection, and (iii) data-quality-aware curriculum learning that stabilizes training. To systematically isolate the impact of this final component, organizing samples based on difficulty and noise, we detail an extensive ablation study in the following section.

6 Ablation Study

To evaluate the impact of curriculum learning strategies in `CHART`, we conduct an extensive ablation study by varying different combinations of difficulty- and noise-based scheduling. The results are presented in Table 5 for both `roberta-base` and `bert-base` models. We observe that the choice of curriculum strategy has a significant impact on model performance across all datasets. In general, strategies that incorporate both difficulty and noise signals outperform those relying on a single factor. This confirms that jointly modeling sample complexity and data quality leads to more effective training. Comparing annealing and expanding strategies, we find that expanding-based curricula (i.e., starting from easier samples and gradually introducing harder ones) tend to perform better on TACRED and AmalREC. For instance, *Difficulty Expanding* achieves strong performance on both datasets. On the other hand, annealing strategies (starting with all samples and gradually removing easier ones) perform competitively on Re-TACRED, suggesting that retaining harder samples in later stages can be beneficial for more challenging datasets.

The best overall performance is achieved by combining noise and difficulty signals in a structured manner. In particular, the *Difficulty Expanding + Noise Annealing* strategy yields the best results for `roberta-base` on TACRED (89.08) and achieves consistently strong performance across all datasets. Similarly, for `bert-base`, the *Noise + Difficulty Annealing* and *Noise Expanding + Difficulty Annealing* strategies produce competitive results, with the latter achieving the best performance on AmalREC (85.31).

From these results, we draw three key insights: (i) combining difficulty and noise is more effective than using either alone,(ii) Expanding curricula help stabilize learning by introducing complexity gradually, and (iii) annealing strategies help the model focus on harder examples in later stages, improving robustness.

Final Strategy Selection. Based on these observations, we adopt the *Difficulty Expanding + Noise Annealing* strategy for `CHART`, as it provides the best overall trade-off across datasets and backbone models. This configuration is used in all main experiments reported in Table 3.

7 Error Analysis

We highlight several types of errors observed in the AmalREC dataset in Table 6. One important characteristic of this dataset is that it does not include a `no_relation` label in its training set, which contributes to certain types of prediction errors. The first common error arises from confusion between semantically similar relation labels, such as `country`, `placeOfBirth`, `country`

Table 5: Ablation results of curriculum learning strategies for `roberta-base` and `bert-base` using CHART (5 epochs). Best results for each model are in bold.

	RoBERTa-base			BERT-base		
Curriculum Strategy	TACRED	Re-TACRED	AmalREC	TACRED	Re-TACRED	AmalREC
Difficulty Annealing	87.78	89.34	85.72	85.38	**89.11**	84.36
Difficulty Expanding	88.89	88.91	**86.56**	86.69	87.40	85.23
Noise Expanding	86.74	87.98	85.04	85.41	87.05	84.61
Noise Annealing	88.65	90.35	86.35	85.98	87.66	84.57
Difficulty + Noise Expanding	83.42	74.69	86.25	84.92	87.43	85.01
Difficulty + Noise Annealing	87.77	90.45	86.42	86.88	87.95	84.62
Noise + Difficulty Annealing	89.03	89.38	85.71	**87.68**	87.91	84.67
Noise + Difficulty Expanding	86.14	90.10	86.25	86.61	87.02	84.35
Noise Annealing + Difficulty Expanding	88.66	**90.93**	85.81	87.10	85.61	85.04
Noise Expanding + Difficulty Annealing	87.06	90.68	86.09	87.43	88.41	**85.31**
Difficulty Annealing + Noise Expanding	86.67	90.36	85.59	84.25	87.26	85.15
Difficulty Expanding + Noise Annealing	**89.08**	89.50	86.26	87.63	88.20	84.22

Table 6: Examples of relation prediction errors in AmalREC. Entity E_1 is shown in red and E_2 in blue.

Sentence	Ground Truth	Predicted
Confusion between similar labels		
Having been born in the picturesque country of France [LOC], Louis Arbessier [PER] spent his formative years absorbing the nation's cultural heritage.	country	placeOfBirth
Tapash Baisya [PER]'s birthplace and citizenship are both Bangladesh [LOC], a country with a rich history.	country	countryOfCitizenship
Predicting parent relation instead of fine-grained relation		
Allan Evridge [PER], a former Kansas State University [ORG] quarterback who won a state championship in the 1980s, later played professional football.	formerHighSchool	highSchool
An NCAA Division I block-leading player, Grady Livingston [PER] studied at Merritt College [ORG] before graduating.	formerHighSchool	educatedAt
Predicting finer-grained relation		
Jon Huntsman Sr. [PER], the father of Jon Huntsman Jr. [PER], passed away on January 3, 2018.	relative	child
Ambiguous relation in sentence		
The picturesque Happy Jack's Creek [LOC], located in the Snowy Mountains [LOC], contributes to the region's scenic beauty.	region	mountainRange

`of citizenship`, `province`, and `region`. Because these labels share overlapping meanings, the model sometimes predicts one label when another closely related label is correct. The second type of error is related to the thresholding strategy used during prediction. We only consider predictions whose scores exceed a predefined threshold. In some cases, the correct label receives the highest score among all labels in that layer but still falls below the threshold, causing the prediction to be discarded. More advanced thresholding techniques could potentially reduce this type of error. The third type of error occurs when the model predicts a more fine-grained relation than the ground truth label. This often happens in sentences describing family or relative relationships, where the directionality of the relation makes a specific relation seem more appropriate. However, the ground truth annotation sometimes uses a broader label such as `relative` to avoid incompatibility between entity types. Finally, errors also arise from ambiguous sentences that contain certain keywords strongly associated with other relation labels. In such cases, the presence of these keywords can mislead the model into predicting an incorrect relation.

8 Conclusion

In this paper, we introduced CHART, a novel framework that addresses the limitations of flat-label classification and the challenges of noisy, varying-complexity datasets in Relation Classification (RC). By reformulating RC as a layer-wise masked prediction task, CHART explicitly models label dependencies using GAT-injected prompt embeddings and a zero-bounded multi-label cross-entropy loss to enforce strict parent-child consistency. To handle heterogeneous data quality, we integrated an LLM-guided curriculum scheduler that automatically organizes training instances based on difficulty and noise. Our comprehensive evaluation across TACRED, Re-TACRED, and the synthetic AmalREC datasets proves that CHART consistently achieves state-of-the-art results against 12 diverse baselines. Furthermore, extensive ablation studies confirm that a joint curriculum strategy of difficulty expanding and noise annealing maximizes model robustness. Ultimately, CHART provides a highly effective, unified approach for extracting consistent hierarchical relations from complex and noisy text

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A LLM Prompt Templates

A.1 Difficulty Judge System Prompt

Prompt for Difficulty Judge System

You are an expert in information extraction and semantic reasoning. Your task is to judge the difficulty of identifying the

correct RDF property (predicate) that relates a subject entity (E1) to an object entity (E2) in a sentence.

Task:
Given a sentence and a list of possible RDF properties, determine how difficult it is to infer the correct property such that:
E1 -> (Property) -> E2
Subject: [Entity A]
Object: [Entity B]

Judgment Criteria:
(1) Clarity: How clearly the sentence supports the correct property.
(2) Reasoning Complexity: How much reasoning is needed to select the most appropriate property among possible choices.
(3) External Knowledge: Whether external/world knowledge is necessary to infer the relation.

Additional Guidelines:
- Minor noise in the sentence is tolerable if the relation remains identifiable.
- The goal is to predict the correct RDF property of E1 whose value is E2.
- Former relationships or associations between E1 and E2 should be considered valid if relevant.

Difficulty Levels:
- Clear
- Contextual
- Implicit
- Knowledge Dependent
- Impossible

Instructions:
(1) Return one of the above difficulty levels.
(2) Provide a short justification for your choice.
(3) Mention overlapping plausible relations if applicable.

A.2 Difficulty Judge User Prompt

Prompt for Difficulty Judge User

Input:
- Sentence: "{sentence}"
- E1: "{E1_entity} [{E1_type}]"
- E2: "{E2_entity} [{E2_type}]"

Gold Sentence:{gold_sentence}
Possible predicates/properties:{possible_predicates}

Output (JSON)

```
{
        "reason": "Your reasoning (max 100 words)"
        "difficulty": "Clear" | "Contextual" | "Implicit" |
        "Knowledge Dependent" | "Impossible"
}
```

A.3 Noise Judge System Prompt

Prompt for Noise Judge System

You are a Noise Judge tasked with evaluating the level of noise in a sentence with respect to a given RDF-style relation tuple: (Subject, Predicate, Object).

Objective:
Determine how well the sentence aligns with the relation tuple and categorize the sentence's quality in terms of noise. Noise is defined as any distortion in grammar, alignment, factuality, or semantic clarity that interferes with identifying or confirming the relation between the subject and object.

Judgment Criteria:
(1) Compatibility: Does the sentence support the relation?
(2) Clarity and Alignment: Are the entities and their relation clearly expressed?
(3) Factual Accuracy: Are there contradictions or misrepresentations?

Additional Guidelines:
- Former relationships between the subject and object should be treated as valid if they support the intended relation.

Noise Levels:
(1) Clean
(2) Minor Noise
(3) Moderate Noise
(4) Severe Noise

Instructions:
- Assign one noise level.
- Provide a short justification referencing the noise type.
- Focus strictly on sentence quality and alignment with the relation.

A.4 Noise Judge User Prompt

Prompt for Noise Judge User

Input:
- Sentence: "{sentence}"
- E1: "{E1_entity} [{E1_type}]"
- E2: "{E2_entity} [{E2_type}]"

Gold Sentence:{gold_sentence}
Possible predicates/properties:{possible_predicates}

Output (JSON)

```
{
    "reason": "Your reasoning (max 100 words)"
    "noise": "Clean" | "Minor Noise" | "Moderate Noise" |
    "Severe Noise"
}
```

B Baseline Implementation

B.1 KnowPrompt [4]

RoBERTa-large was used as the primary backbone. The batch size is kept at 16, and gradient accumulation occurs after 4 steps. We utilized a maximum sequence length of 256 and a regularization coefficient $t_\lambda = 0.001$. Number of epochs was kept as 5 with a learning rate of 3×10^{-5}. WIKI80 is used as the data class for all three datasets.

B.2 GenPT [8]

RoBERTa-large was used as the primary backbone for all experiments. We set the per-device training batch size to 4 for AmalRec and 16 for TACRED and Re-TACRED, with gradient accumulation steps fixed at 1. The maximum sequence length was set to 256 for AmalRec and 501 for TACRED and Re-TACRED. The maximum entity type length and label length were configured to 5 and 11 for AmalRec, and 4 and 6 for TACRED-based datasets, respectively. We trained the models using the Adam optimizer with a learning rate of 3×10^{-5}, while newly introduced tokens were assigned a smaller learning rate of 1×10^{-5}. A linear warmup of 10 steps was applied, and gradient clipping was performed with a maximum norm of 1. The number of training epochs was set to 5 for AmalRec and 10 for TACRED and Re-TACRED. Experiments were conducted under standard settings using fixed random seeds.

B.3 PTR [10]

RoBERTa-large was adopted as the backbone model for all experiments. The per-device training batch size was set to 8 with gradient accumulation steps of 1. A maximum sequence length of 512 was used across all datasets. The models were optimized using Adam with a base learning rate of 3×10^{-5}, while newly introduced tokens were assigned a smaller learning rate of 1×10^{-5}. We applied a warmup of 500 steps, along with weight decay of 1×10^{-2} and an Adam epsilon value of 1×10^{-6} for numerical stability. The number of training epochs was fixed at 5. For each dataset (AmalRec, Re-TACRED, and TACRED), separate directories for train, validation, and test splits were ensured before training, and prompt templates were provided via an external file (`temp.txt`). We handcrafted the prompt templates for the AmalREC dataset, while those for TACRED and Re-TACRED were provided in the respective GitHub repositories.

B.4 HPT [34]

Same configuration as provided in the implementation setup. We train for 5 epochs. Base model is kept as `bert-base-uncased`.

B.5 AGGCN [7]

Pre-trained word embeddings are created of size 300. The POS and NER embedding dimensions are set to 30 each, and the hidden size is fixed at 300 with 2 GCN layers and 3 attention heads. Dropout rates are set to 0.5 for input, GCN, and RNN layers, with an additional word dropout of 0.04. Max pooling is used with an L2 regularization coefficient of 2×10^{-3}, and a single-layer MLP is applied for classification. The model is trained using stochastic gradient descent (SGD) with an initial learning rate of 0.5, which is decayed by a factor of 0.9 after 5 epochs. Training is conducted for 50 epochs with a batch size of 32, and gradient clipping is applied with a maximum norm of 5. This is the same as the default parameters provided on GitHub.

B.6 GCN over pruned trees [38]

Pre-trained word embeddings of dimension 300 are created, along with POS and NER embeddings of size 30 each. The hidden dimension is set to 200, and the model uses 2 GCN layers without an additional RNN encoder. Dropout is applied with rates of 0.5 for both input and GCN layers, and a word dropout of 0.04 is used. Max pooling is adopted, followed by a 2-layer MLP classifier, with an L2 regularization coefficient of 3×10^{-3} applied to the pooled representation. Dependency tree pruning is enabled with $k = 1$. The model is trained using stochastic gradient descent (SGD) with a learning rate of 0.3, decayed by a factor of 0.9 after 5 epochs when performance saturates. Training is conducted for 100 epochs with a batch size of 50, and gradient clipping is applied with a maximum norm of 5. This is the same as the default parameters provided on GitHub.

B.7 LLM Prompting [1, 6, 11, 32, 36]

We perform inference using open-weight large language models accessed via the Hugging Face Transformers library. Specifically, we use `<model_id>` (e.g., LLaMA, Mistral, or similar instruction-tuned variants), loaded in bfloat16 precision with automatic device mapping. All models are used in a zero-temperature (greedy decoding) setting with a maximum generation length of 20 tokens. We follow the model-specific usage guidelines as provided in their respective Hugging Face model cards. We employ a constrained few-shot prompting strategy where only entity-type-compatible examples are selected. For each test instance, candidate relations are restricted based on the entity type pair, and $k = 5$ few-shot demonstrations are dynamically retrieved from the same entity-type combination. The model is prompted to select the correct relation label from the provided candidate set using greedy decoding.

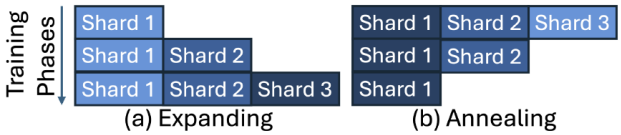

Figure 4: Various scheduling strategies in curriculum learning. Expanding adds difficult samples in subsequent phases whereas annealing removes easier samples in subsequent phases.

C Figures and Tables

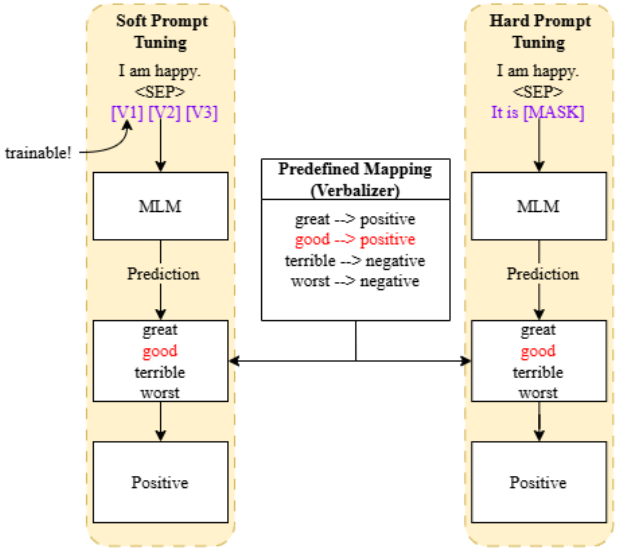

Figure 5: Difference in Hard and Soft Prompt Tuning

Table 8: Distribution of relation instances across difficulty and noise levels for each dataset. Values are percentages.

Level	TACRED	Re-TACRED	AmalREC
Difficulty Levels			
Clear	98.83	98.98	99.96
Contextual	0.92	0.86	0.03
Implicit	0.17	0.10	0.01
Impossible	0.08	0.06	0.00
Noise Levels			
Clean	53.14	57.36	85.80
Minor Noise	36.64	34.19	13.77
Moderate Noise	8.97	7.18	0.42
Severe Noise	1.25	1.27	0.01

Table 10: Average and maximum performance (Avg/Max) of different backbone models across curriculum learning strategies using CHART (5 epochs).

Model	TACRED	Re-TACRED	AmalREC
`roberta-base`	87.49 / 89.08	88.56 / 90.93	86.00 / 86.56
`bert-base-uncased`	86.33 / 87.68	87.58 / 89.11	84.76 / 85.31

D Illustrative Example of the Proposed Framework CHART

In this section, we provide a step-by-step illustration of our framework using a concrete example to demonstrate how each component operates.

Input and Problem Setup

Consider the input sentence:

> *"Within Louisiana, a state that serves as both the residence and the home for John Henry Baker, a leader who currently lives and fosters deep connections with the local community, the high walls and security guards surrounding his place of residence emphasize the notable presence of John Henry Baker."*

Let the entity pair be:

$$(E_1, E_{1,\text{type}}) = (\text{John Henry Baker, Person}),$$
$$(E_2, E_{2,\text{type}}) = (\text{Louisiana, Location})$$

Assume a relation hierarchy with two levels:

- `Person-OrganizationAffiliation`
  - `employer`
  - `awardedBy`
  - `agency`
- `person:geographicConnection`
  - `residence`
  - `restingPlace`
  - `locatedAt`

The ground truth relation is:

$$y = \texttt{residence}$$

LLM-based Difficulty and Noise Estimation

The input sentence is passed to a large language model with a structured prompt to estimate its difficulty and noise level.

Output:

- Difficulty: `Clear`
- Noise: `Clean`

This classification is later used for curriculum scheduling.

D.1 Curriculum Scheduling

Based on the assigned difficulty, this sample is placed in the `Easy` shard.

During training:

- In the **expanding strategy**, this sample is introduced in the initial phase.
- In the **annealing strategy**, it may be removed in later phases to focus on harder examples.

Table 7: Model Pair Agreement Comparison for Noise

Model Pair Comparison	Agreement Score	# Disagreement Cases	Agreement (%)
Llama-3.1-8B-Instruct ↔ Llama-3.2-3B-Instruct	0.682	122	68.2%
Llama-3.1-8B-Instruct ↔ SmolLM3-3B	0.781	84	78.1%
Llama-3.1-8B-Instruct ↔ Mistral-7B-Instruct-v0.2	0.771	88	77.1%
Llama-3.1-8B-Instruct ↔ Zephyr-7B-Beta	0.776	86	77.6%
Llama-3.2-3B-Instruct ↔ SmolLM3-3B	0.792	80	79.2%
Llama-3.2-3B-Instruct ↔ Mistral-7B-Instruct-v0.2	0.786	82	78.6%
Llama-3.2-3B-Instruct ↔ Zephyr-7B-Beta	0.797	78	79.7%
SmolLM3-3B ↔ Mistral-7B-Instruct-v0.2	0.958	16	95.8%
SmolLM3-3B ↔ Zephyr-7B-Beta	0.953	18	95.3%
Mistral-7B-Instruct-v0.2 ↔ Zephyr-7B-Beta	0.966	13	96.6%

Table 9: Examples of difficulty and noise prediction by Llama-3.2-3B-Instruct in AmalREC. Entity E_1 is shown in red and E_2 in blue.

Sentence	Relation	Difficulty / Noise
Given Examples		
Within Louisiana [LOC], a state that serves as both the residence and the home for John Henry Baker [PER], a leader who currently lives and fosters deep connections with the local community, the high walls and security guards surrounding his place of residence emphasize his notable presence.	residence	Clear / Clean
The Porcovita River [LOC], a renowned hiking destination and Olt River tributary, meanders gracefully from its source in the majestic, peak-filled Lotru Mountains [LOC], serving as a vital lifeline while originating within their picturesque landscape.	sourceMountain	Contextual / Minor Noise
The distinguished Matilda of Bethune [PER], countess of Flanders and a figure of historical prominence, found her final resting place in Flines-lez-Raches [LOC], a serene location in France where she was peacefully buried after her demise.	placeOfBurial	Implicit / Clean
Drawing inspiration from and collaborating with Neil Young [PER], a versatile and legendary musical artist, Lukas Nelson & Promise of the Real [ORG] have been making waves as a rising musical organization.	musicalArtist	Implicit / Minor Noise
While showcasing natural beauty, Gader Valley [LOC] is nestled within the heart of the Zilina Region [LOC], a region of Slovakia.	region	Impossible / Clean
Nestled within the Swiss Alps, Kander (Switzerland) [LOC] functions as the river's meeting point with Switzerland [LOC], thereby serving as a breathtaking mouthpiece showcasing its natural beauty.	mouthCountry	Clear / Moderate Noise
Though John Bosco [PER] was beatified to heaven by the honorable Pope Pius XI [PER], it was Pope Pius XI who beatified John Melchior Bosco on 2 June 1929.	beatifiedBy	Clear / Severe Noise

Table 11: Performance comparison of curriculum strategies (Annealing vs. Expanding) across backbone models using CHART (5 epochs).

Strategy	Model	TACRED	Re-TACRED	AmalREC
Annealing	`roberta-base`	88.09	90.12	85.99
	`bert-base-uncased`	86.54	87.76	84.74
	Average	**87.31**	**88.94**	85.37
Expanding	`roberta-base`	87.08	87.89	85.98
	`bert-base-uncased`	86.26	87.30	84.87
	Average	86.67	87.60	**85.42**

D.2 Hierarchy-aware Prompt Construction

The input is converted into a prompt-based format:

> [CLS] Within Louisiana, a state that serves as both the residence and the home for John Henry Baker, a leader who currently lives and fosters deep connections with the local community, the high walls and security guards surrounding his place of residence emphasize the notable presence of John Henry Baker. Which predicate of John Henry Baker [PERSON] has value Louisiana [LOCATION]? [V1] [PRED] [V2] [PRED] [SEP]

Here:

- V_1, V_2 are learnable virtual tokens
- Each `[PRED]` corresponds to one hierarchy level

D.3 Model Encoding

The prompt is tokenized and passed through the encoder, producing hidden representations:

$$H = [h_1, \ldots, h_N, h_{V_1}, h_{P_1}, h_{V_2}, h_{P_2}]$$

where h_{P_1} and h_{P_2} correspond to prediction slots for Level 1 and Level 2 respectively.

D.4 Hierarchy Injection via Graph Encoding

A graph is constructed over relation labels:

- `Person-OrganizationAffiliation`
  - `employer`
  - `awardedBy`
  - `agency`
- `person:geographicConnection`
  - `residence`
  - `restingPlace`
  - `locatedAt`

Virtual nodes corresponding to each level are connected to label nodes. Graph propagation updates the prompt embeddings:

$$t_i' = t_i + g_i$$

This allows the prompt to encode hierarchical dependencies.

D.5 Layer-wise Prediction

The model produces scores for each label:

- Level 1 prediction (from h_{P_1}):

$$\texttt{Person-OrganizationAffiliation} : -0.9$$
$$\texttt{person:geographicConnection} : 1.7$$

- Level 2 prediction (from h_{P_2}):

$$\begin{aligned} \texttt{employer} &: -2.3 \\ \texttt{awardedBy} &: -1.1 \\ \texttt{agency} &: -0.6 \\ \texttt{residence} &: 2.7 \\ \texttt{restingPlace} &: 1.1 \\ \texttt{locatedAt} &: 0.8 \end{aligned}$$

D.6 Loss Computation

Using zero-bounded multi-label cross entropy:

- Positive labels are encouraged to have scores > 0
- Negative labels are pushed below 0

Loss is computed independently for each hierarchy level and combined with masked language modeling loss.

D.7 Inference with Thresholding

During inference, a threshold τ (learned via cross-validation) is applied.

Assume $\tau = 0$:

- Level 1: `person:geographicConnection` selected
- Level 2: `residence` selected

D.8 Final Prediction

The final predicted relation is:

$$\texttt{person:geographicConnection} \rightarrow \texttt{residence}$$

This matches the ground truth, demonstrating the effectiveness of hierarchical modeling and structured prompting.

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